

Current Position

Postdoctoral Research Fellow Carleton University Supervisor: Millard K 2026–

Work Experience

Research Associate Carleton University Supervisor: Millard K 2025–2026

Education

Ph.D	University of Toronto	2020–2026
	Supervisor: Fortin M-J, James PMA	
M.Sc.	Concordia University	2017–2019
	Supervisor: Brown GE	
B.Sc (Hons)	University of Toronto	2013–2017
	Supervisor: Malcolm JR	

Publications

In Preparation

Millard K, Goldman JA*, Pelletier N, Thompson DK, Pontone N, Guindon L. Severe droughts are putting Canadian peatland carbon stocks at risk: an analysis of fire selectivity and behavior in peatlands over the extreme 2023 fire season.
Wilkinson SL, Leverkus S, Goldman JA*, Butchart C, Flannigan M. Multi-year overwintering fires in Boreal Canada.
Wilkinson SL, Goldman JA*, Krieger-Pottruff E, James PMA. Legacies of mountain pine beetle defoliation influences burn severity in western Canada
Goldman JA*, Fortin MJ, James PMA. The impact of spruce budworm defoliation legacies on wildfire burn severity and post-fire recovery.

In Review

Goldman JA*, Fortin MJ, James PMA. Within and among year moisture availability influences wildfire burn severity in the boreal forest. *Fire Ecology*.

Published

7. Goldman JA*, Fortin MJ, James PMA. 2025. Snow dynamics and forest structure interact to increase wildfire burn severity in the boreal forest. *Forest Ecology and Management*.
6. Messaoud Y, Reid A, Tchebakova NM, Goldman JA*, & Hofgaard A. 2022. The historical complexity of tree height growth dynamic associated with climate change in western north america. *Forests*, 13(5): 738.
5. Crane AL, Bairos-Novak KR, Goldman JA*, Brown GE. 2022. Chemical disturbance cues in aquatic systems: a review and prospectus. *Ecological Monographs*, 92(1): e01487.
4. Goldman JA*, Crane AL, Feyten LEA, Collins E, Brown GE. 2022. Disturbance cue communication is shaped by emitter diet and receiver background risk in Trinidadian guppies. *Current Zoology*, 68(4): 433440.
3. Brown GE, Demers EM, **Goldman JA*, Singh A, Chivers DP, Ferrari MCO. 2020. Unpredictable risk enhances induced neophobia in northern red-bellied dace. *Animal Behaviour*, 168: 121127.
2. Goldman JA*, Desormeaux IS, Brown GE. 2020. Disturbance cues function as sources of risk assessment information under natural conditions. *Freshwater Biology*, 65(5): 981986.
1. Goldman JA*, Feyten LEA, Ramnarine IW, Brown GE. 2020. Sender and receiver experience with risk alters the response to disturbance cues by Trinidadian guppies. *Current Zoology*, 66(3): 255261.

Presentations

2026	The impact of spruce budworm defoliation legacies on wildfire burn severity and post-fire recovery. Wildfire Canada
2024	Comparing forest recovery trajectories following wildfire in the presence and absence of spruce budworm defoliation using remotely sensed data. IALE-North America Annual Meeting, Oklahoma City, Oklahoma.
2023	The effects of snowmelt timing on wildfire burn severity in the boreal forest. IALE-North America Annual Meeting, Riverside, California.
2022	Time-lagged and cumulative effects of moisture availability on burn severity in the boreal forest. Wildland Fire Canada Conference, Edmonton, AB.
2022	Trends in burn severity across an east-west gradient. Ecological Society of America Annual Meeting, Montreal, QC.
2022	Spatial and temporal trends in landscape patterns of burn severity in the boreal shield ecozone of Ontario. North American Forest Ecology Workshop (virtual).

Scholarships & Awards

2024-2025	School of Graduate Studies Doctoral Completion Award
2023-2024	Queen Elizabeth II Graduate Scholarship in Science and Technology
2022	North American Forest Ecology Workshop Student Presentation Award 3rd Place
2020-2021	Edward H. Buckley and Grace B. Buckley Graduate Scholarship in Forestry
2020-2021	GSEF Faculty of Forestry Award
2020-2021	Louis J. Nozzolillo Fellowship in Forestry Research
2020-2023	NSERC Postgraduate Scholarship Doctoral

Teaching

2023	FOR3012: Analytical Methods in Forestry Statistics (graduate) Teaching Assistant	University of Toronto
2020	BIO120: Adaptation and Biodiversity Teaching Assistant	University of Toronto
2019	BIOL 4259/9915: Research Hypothesis Testing (undergrad/grad) Teaching Assistant	Concordia University
2018	BIOL 227: Laboratory Studies of Organisms Teaching Assistant	Concordia University
2017	BIOL 201: Introductory Biology Teaching Assistant	Concordia University

Lectures & Workshops

2025	Harnessing The Power of Google Earth Engine with Python University of Toronto
2023	Multivariate Analysis and Ordination University of Toronto

Professional Development

2023	Short Course on Forecasting for Decision-Making: An Epidemiological & Ecological Perspective (Invited)
2023	Canadian Ecological Forecasting Initiative & Fields Institute, University of Toronto Structural Equation Modelling Summer School Université de Sherbrooke

Skills & Tools

Programming: R, Python, JavaScript, bash

Project Management: git, uv, uvr, Quarto

Data Management: duckDB, PostgreSQL

GIS: Google Earth Engine, QGIS, ArcPy